Industrial Pollution is Putting Kids at Risk

Exposure to toxic chemicals and particulate matter produced by activities like manufacturing, waste management, chemical production, and gas processing has been shown to increase the risk of cardiovascular, respiratory, and other diseases, including cancer. These impacts are especially pronounced among vulnerable populations like children. While zoning laws in most of the U.S. restrict industrial facilities to designated areas, these areas are often still located in close proximity to schools. Economically and racially marginalized communities are disproportionately affected by this phenomenon, compounding existing health inequities. Despite some environmental regulations in the United States, industry continues to profit while exposing vulnerable groups to hazardous pollution.

Most at-risk schools

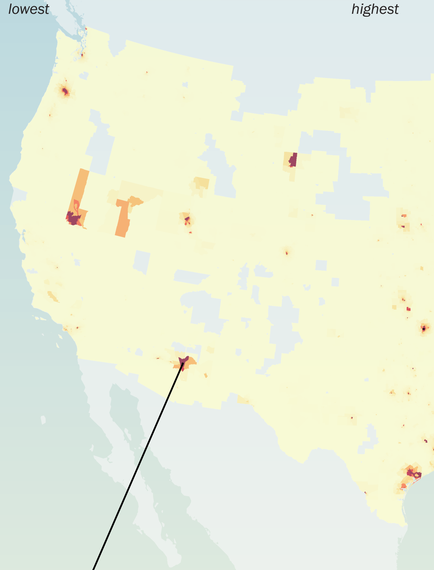
The EPA's Risk- Screening Environmental Indicator (RSEI) dataset maps air pollution across the U.S. using a measure called "toxicity concentration." This value represents the sum of pollutant concentration an area is exposed to, weighted by a measure of inhalation toxicity by chemical.

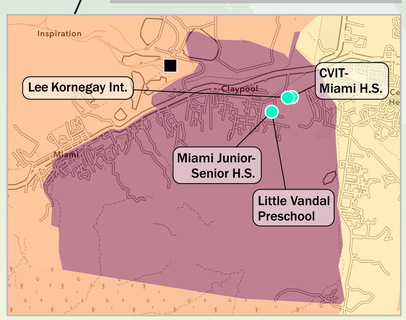
The following maps show the schools with the highest underlying toxicity concentrations. In many cases, schools are located essentially across the street from industrial facilities, exposing children to huge amounts of dangerous air pollution.

Toxicity Concentration

lowest — *highest*

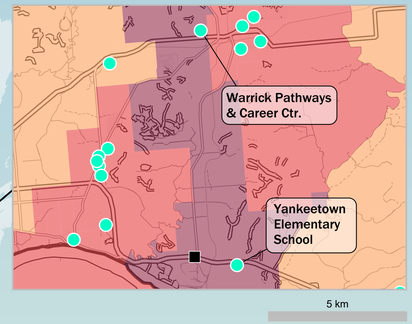
Public school ■ Industrial facility



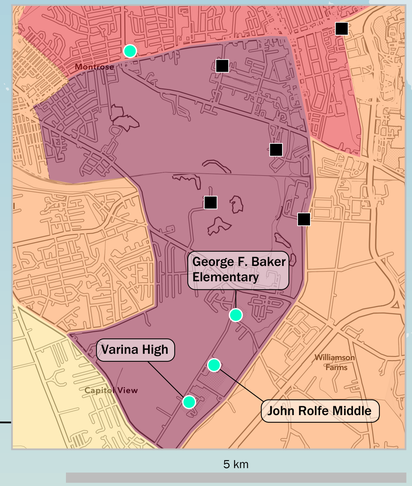


Miami, Arizona

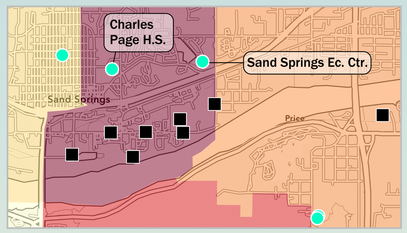
Newburgh & Boonville, Indiana



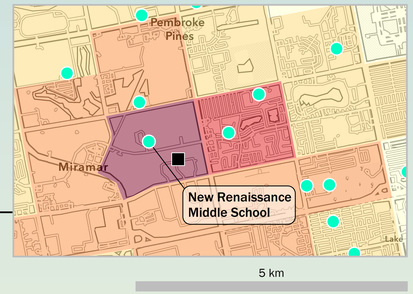
Richmond, Virginia

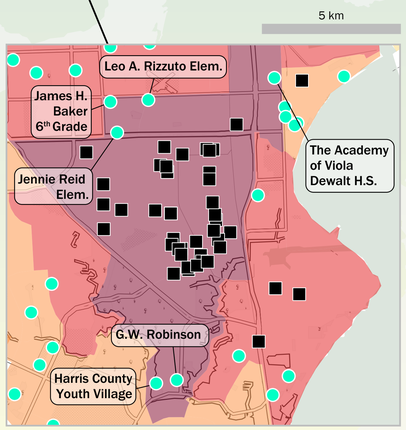


Sand Springs, Oklahoma

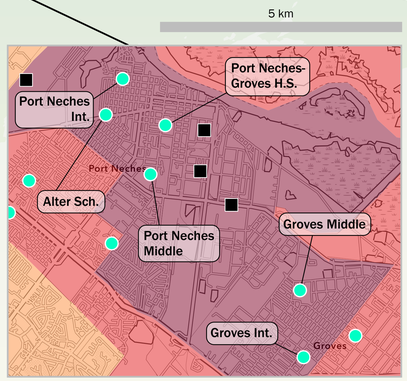


Miramar, Florida

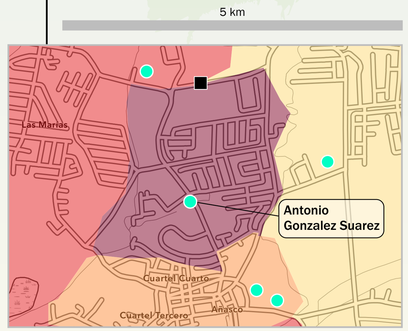




La Porte & Seabrook, Texas



Groves & Port Neches, Texas



Añasco, Puerto Rico

Worst offenders

These are the highest emitting industrial facilities and parent companies located within five kilometers of a school.

Facility	Location	Emissions	Sector
Freeport-McMoran Miami	Claypool, AZ	14k tons	Primary Metals
Ascend Performance Materials - Pensacola Plant	Cantonment, FL	10k tons	Chemicals
Kennecott Utah Copper Smelter & Refinery	Magna, UT	10k tons	Primary Metals
Nyrstar Clarksville	Clarksville, TN	8k tons	Metal Mining
Lucky Friday Mine	Mullan, ID	8k tons	Metal Mining
Bayer CropScience - Luling Site	Luling, LA	8k tons	Chemicals
AK Steel Rockport Works	Rockport, IN	7k tons	Primary Metals
Chemours Delisle Plant	Pass Christian, MS	7k tons	Chemicals
Tronox, Hamilton Facility	Hamilton, MS	7k tons	Chemicals
Oil Technology/U.S. Steel Gary	Gary, IN	6k tons	Primary Metals

Owner	Emissions	Facilities	Share of Owner Emissions	Share of Owner Facilities
Freeport-McMoran	18k tons	4	74%	33%
Cleveland-Cliffs	15k tons	29	89%	97%
Nyrstar US	13k tons	6	100%	100%
US Department of Defense	12k tons	222	81%	75%
International Paper	11k tons	17	54%	61%
The Chemours Co.	11k tons	13	98%	72%
US Steel	11k tons	16	100%	94%
Ascend Performance Materials	11k tons	4	36%	67%
Bayer US	10k tons	4	100%	80%
Rio Tinto America	10k tons	2	13%	67%

Proximity modeling

School exposure to industrial facilities impacts low income and racial minority communities most. The mechanisms behind this trend are clear: wealthier and more privileged residents can afford to live in less polluted areas, and communities with more economic and social power are more able to push back against new industrial development in their towns and neighborhoods through strict zoning laws.

The following are the results of a logistic model relating whether a school is within five kilometers of an industrial facility to underlying economic and demographic characteristics of the area.

An increase of	changes the odds by
5 years in median age	0.78x
$10,000 in median income	0.92x
10% in Black residents	1.09x
10% in white residents	0.84x
10% in the unemployment rate	0.96x
10% in the share below 150% of poverty line	1.07x

Data sources

- 2021–2022 National Center for Education Statistics' Public Schools dataset
- 2022 EPA Toxics Release Inventory (TRI) and EPA Facility Registry Service (FRS)
- 2022 EPA Risk-Screening Environmental Indicator (RSEI) model, census-tract level
- 2021 American Community Survey (ACS) via Census API

Data processed in R and Python using `arcpy`. Maps created in ArcGIS Pro.

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